

Déploiement OpenVPN

Kompta Plus Bien (KPB) — Solution VPN Sécurisée

2 VMs Debian • OpenVPN 2.6 • AES-256-GCM • PKI easy-rsa

Contexte & Objectif

🏢 Client — KPB

Kompta Plus Bien (KPB)
Cabinet comptable — Marignane
~30 salariés en télétravail

Besoin : accès sécurisé aux documents comptabilité, factures & dossiers partagés

🎯 Objectifs du projet

- Installer OpenVPN sur Debian
- PKI complète via easy-rsa
- Chiffrement fort AES-256-GCM
- Maquette 2 VMs (VMware)
- Résilience si VPN tombe
- Bonus : 2FA Google Auth

Architecture du Lab

VM1 — SERVEUR

Debian 12 Bookworm
OpenVPN 2.6.14
IP : 172.16.228.133
tun0 : 10.8.0.1
Port UDP 1194
easy-rsa PKI
CA : KPB-CA

Tunnel VPN
AES-256-GCM
UDP 1194

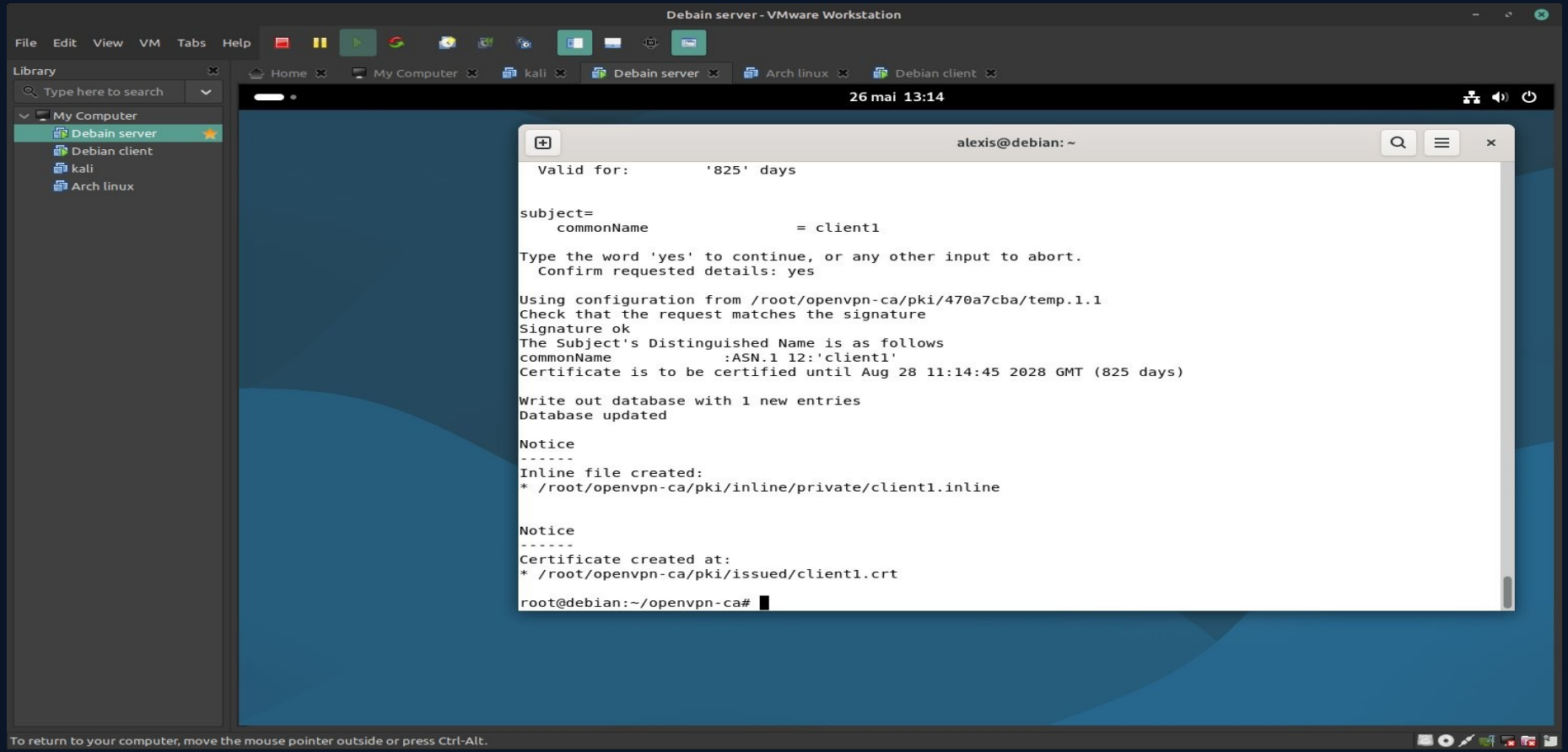
VM2 — CLIENT

Debian 12 Bookworm
OpenVPN Client
IP : 172.16.228.136
tun0 : 10.8.0.6
Cron Watchdog
Auto-restart VPN
systemctl enable

PKI — Étape 5

Génération des certificats

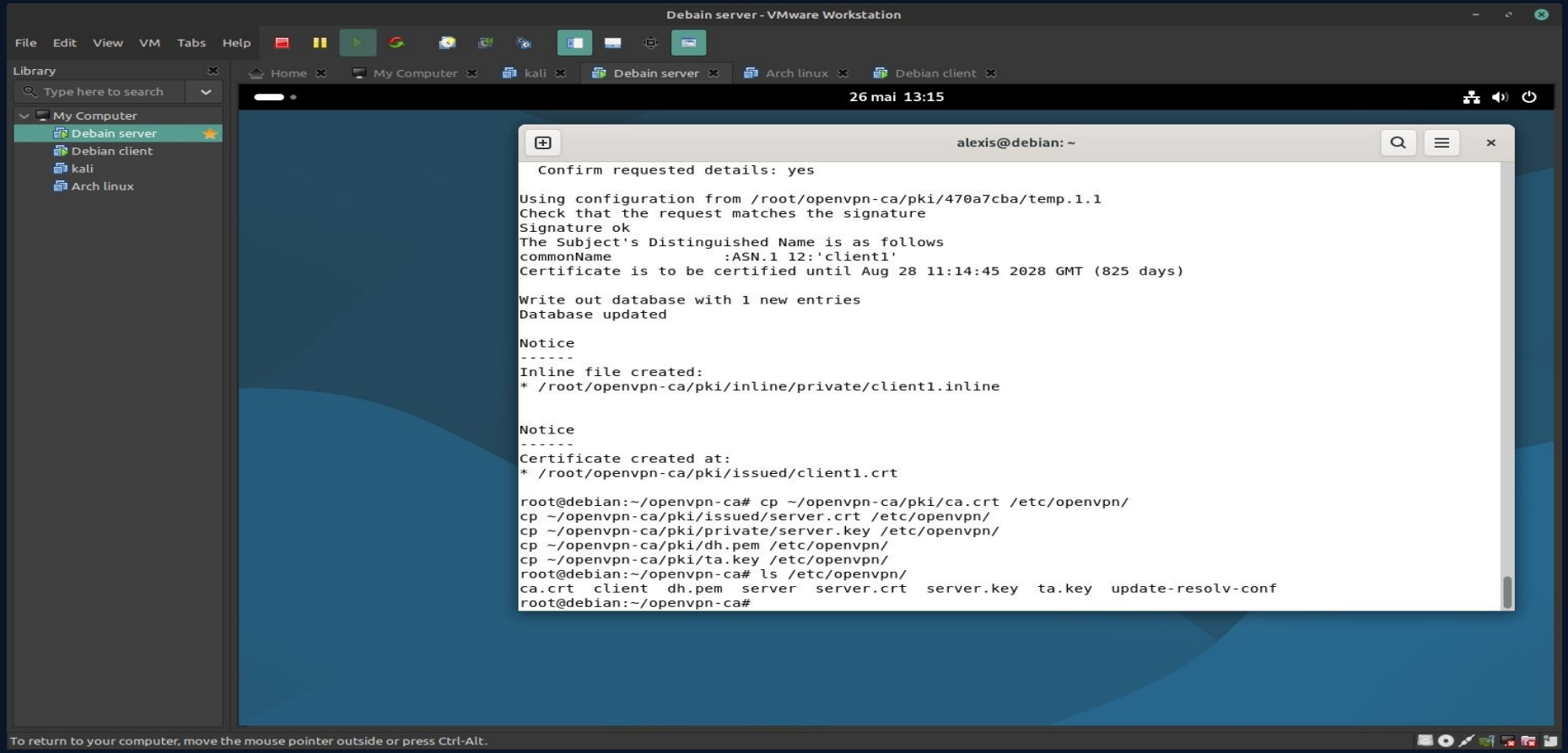
Certificat client1 signé — valid until Aug 2028



Serveur — Étape 8

Copie des fichiers PKI

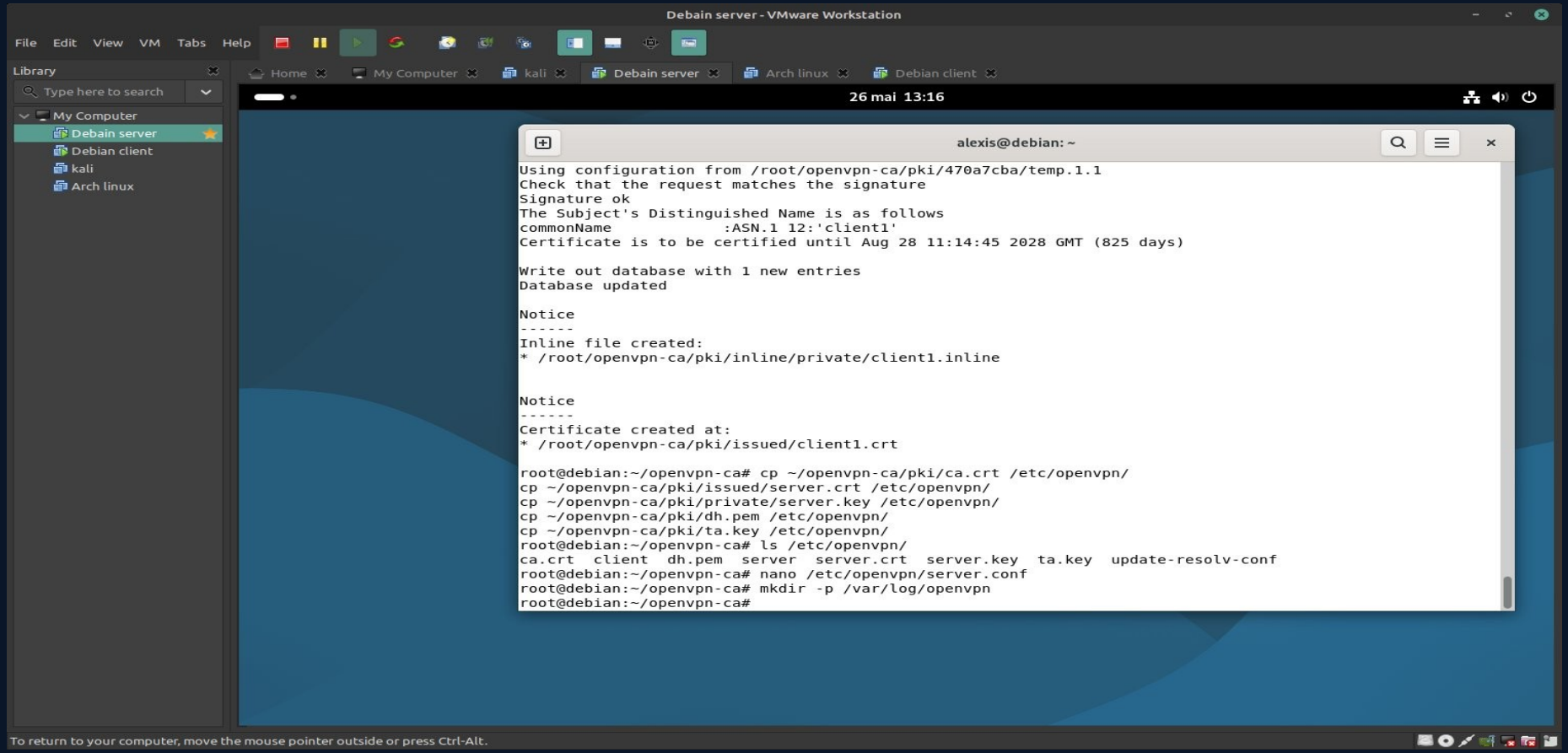
ca.crt, server.crt, server.key, dh.pem, ta.key → /etc/openvpn/



Serveur — Étape 9

Configuration serveur

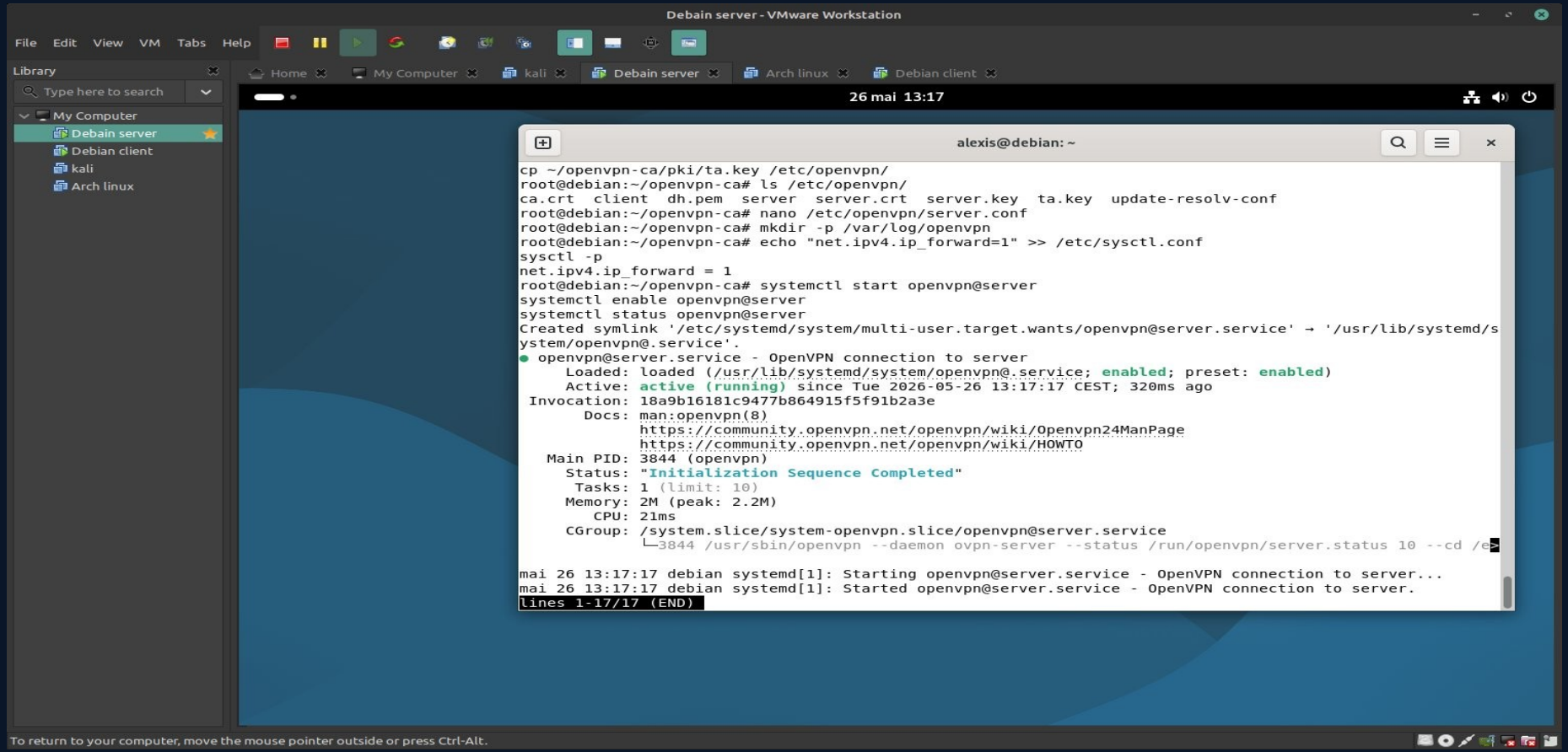
server.conf créé + dossier /var/log/openvpn/



Serveur — Étape 10

Serveur OpenVPN actif

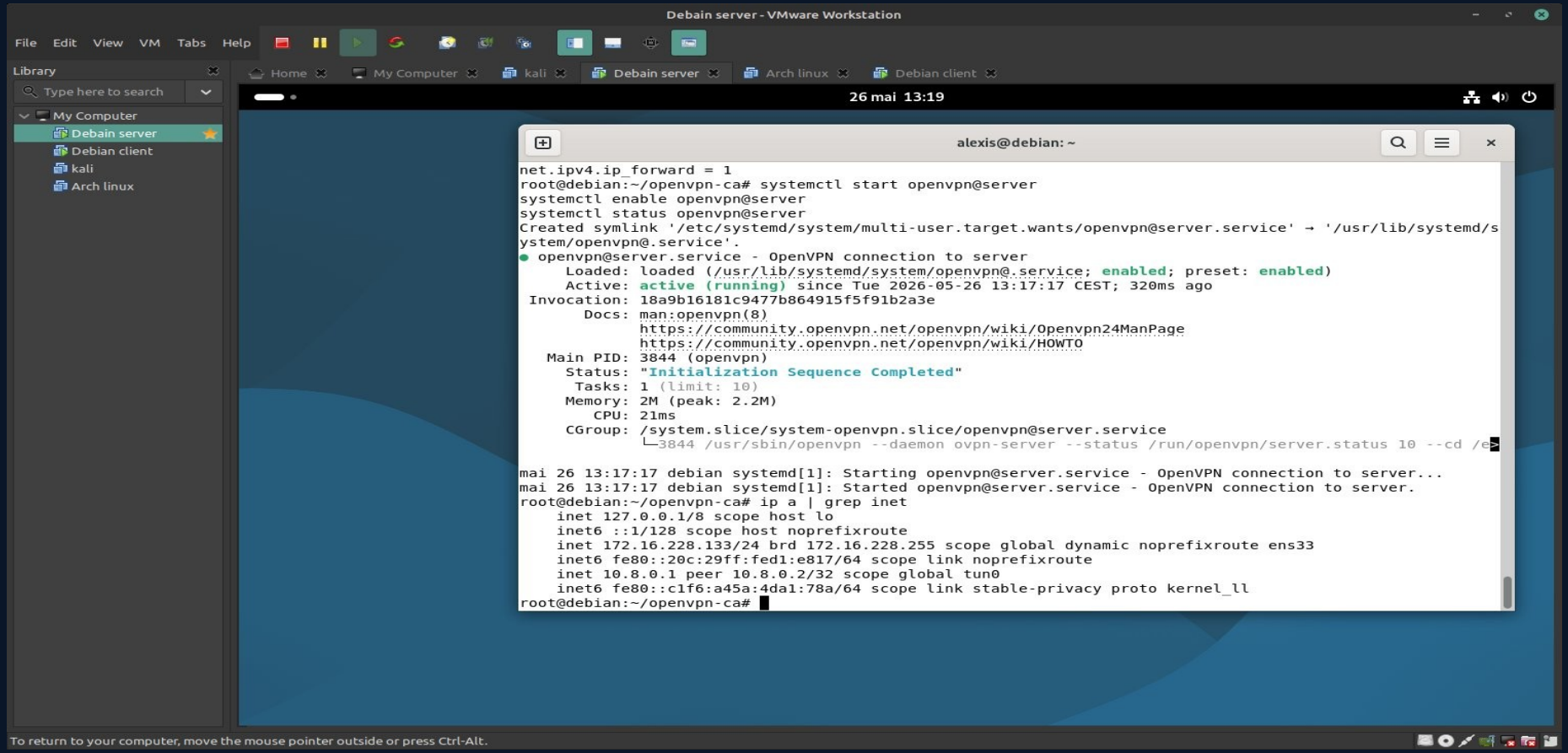
active (running) — Initialization Sequence Completed



Réseau — VM1

Interface tun0 visible

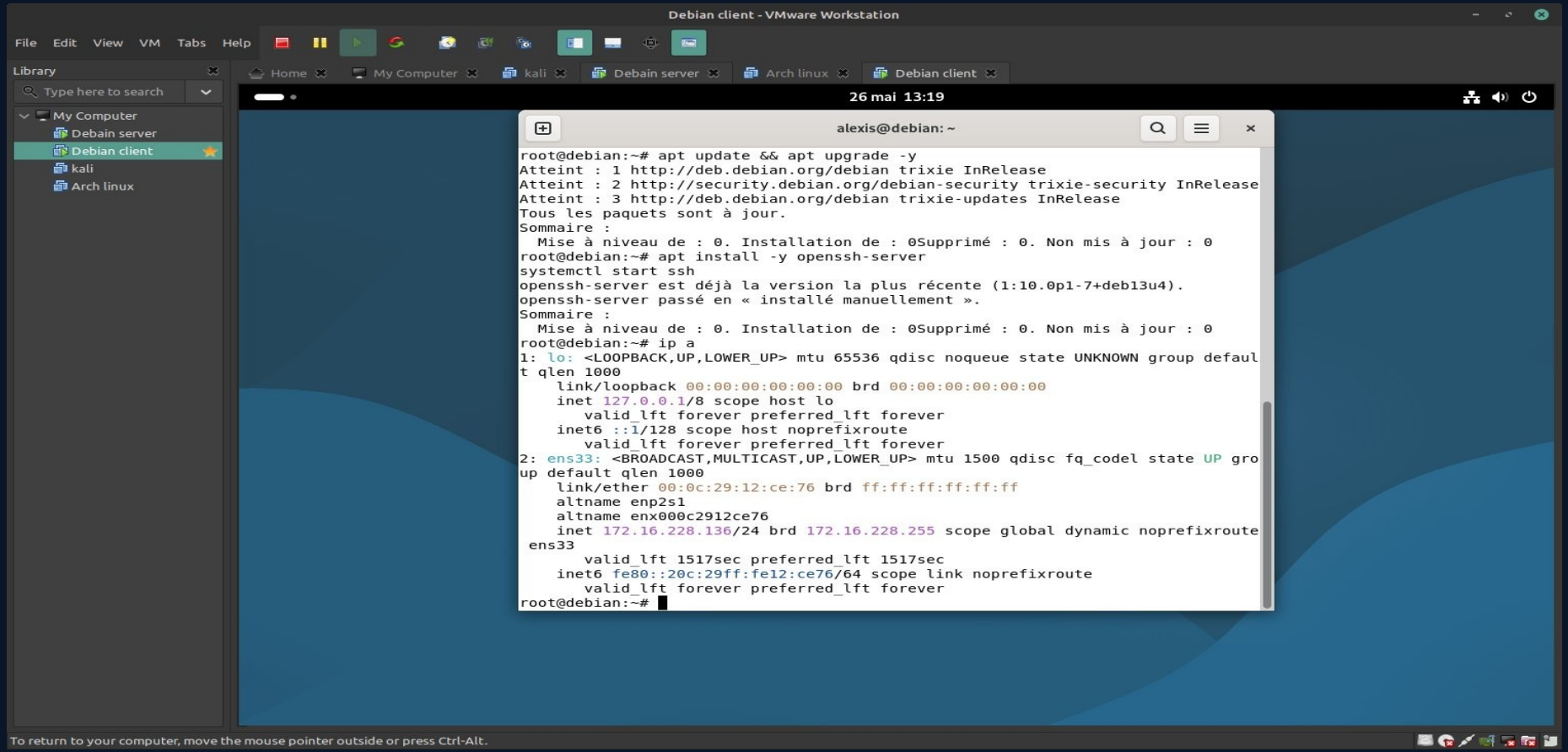
VM1 : 172.16.228.133 — tun0 : 10.8.0.1



Réseau — VM2

IP du client

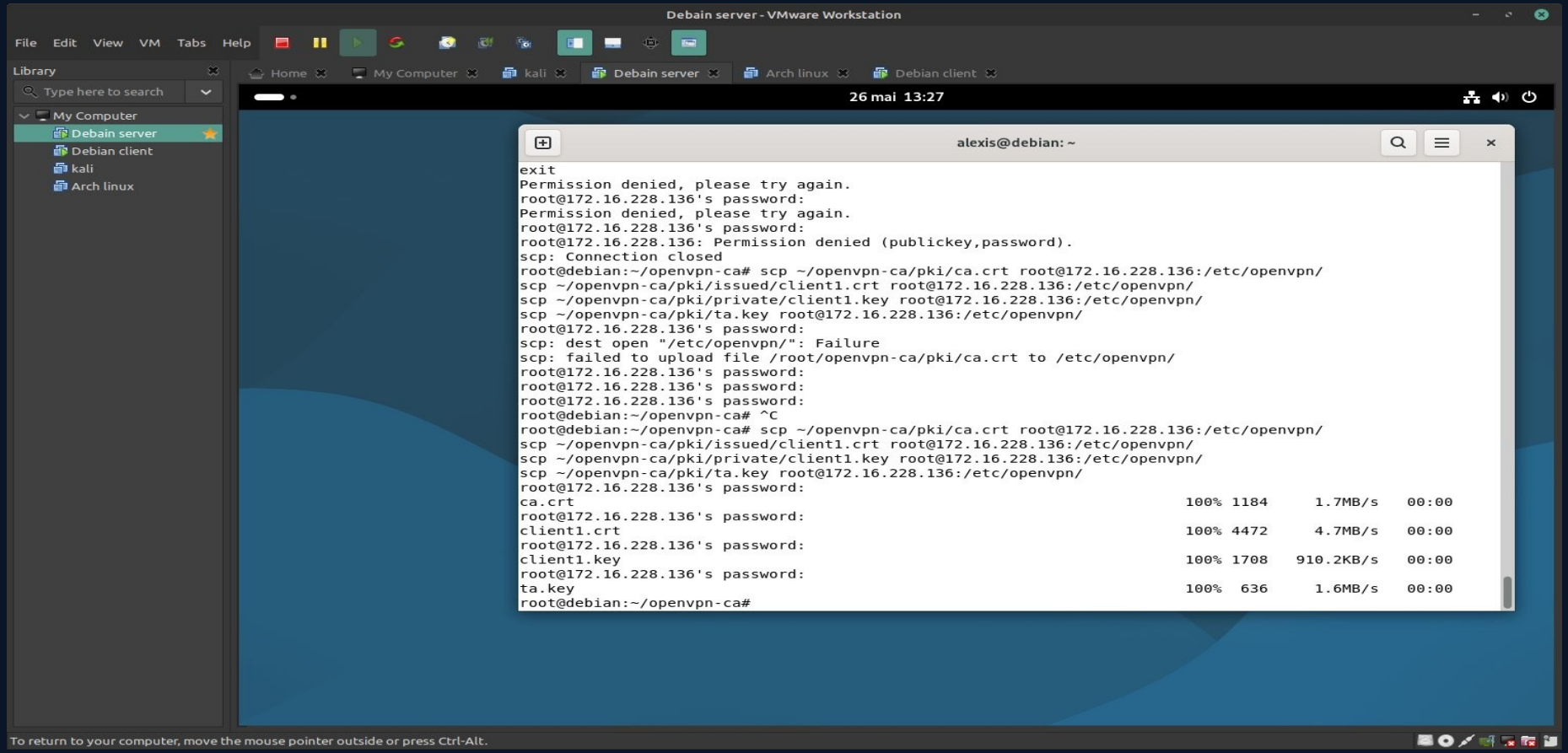
VM2 : 172.16.228.136 — SSH activé



Client — Étape 11

Transfert SCP réussi

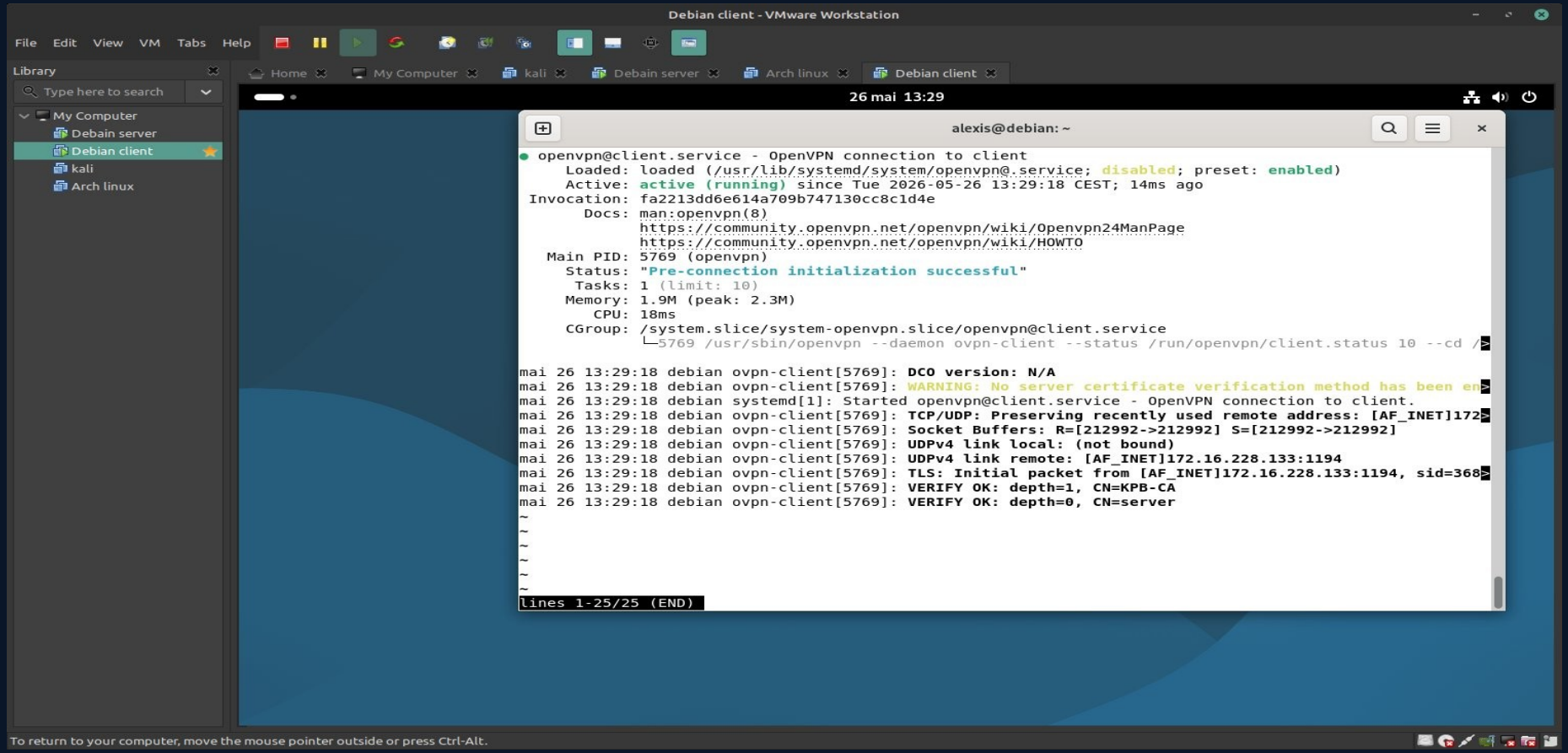
ca.crt, client1.crt, client1.key, ta.key → 100%



Client — Étape 12

Client OpenVPN — pré-connexion

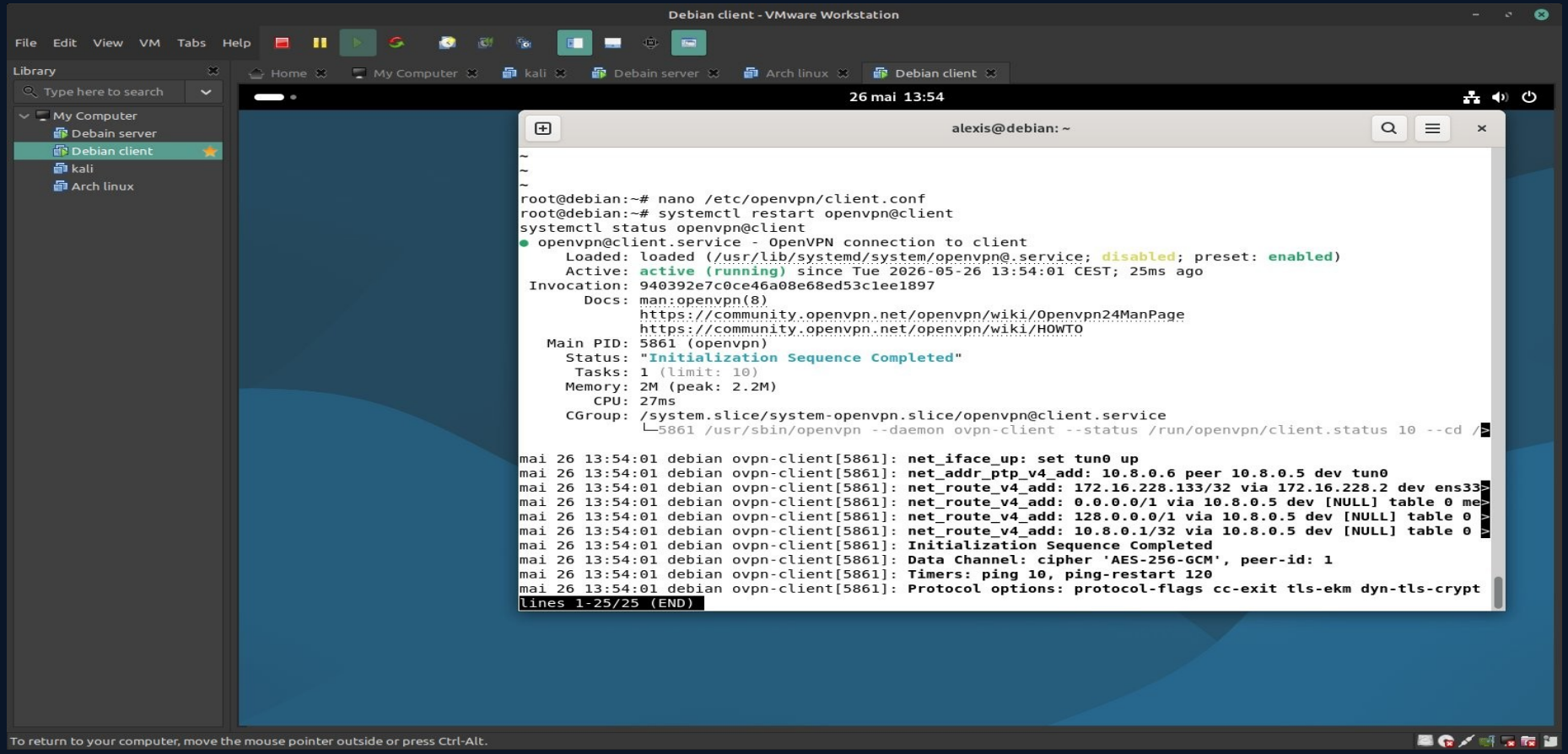
VERIFY OK: KPB-CA — VERIFY OK: server



Tunnel — ✅

Tunnel VPN établi

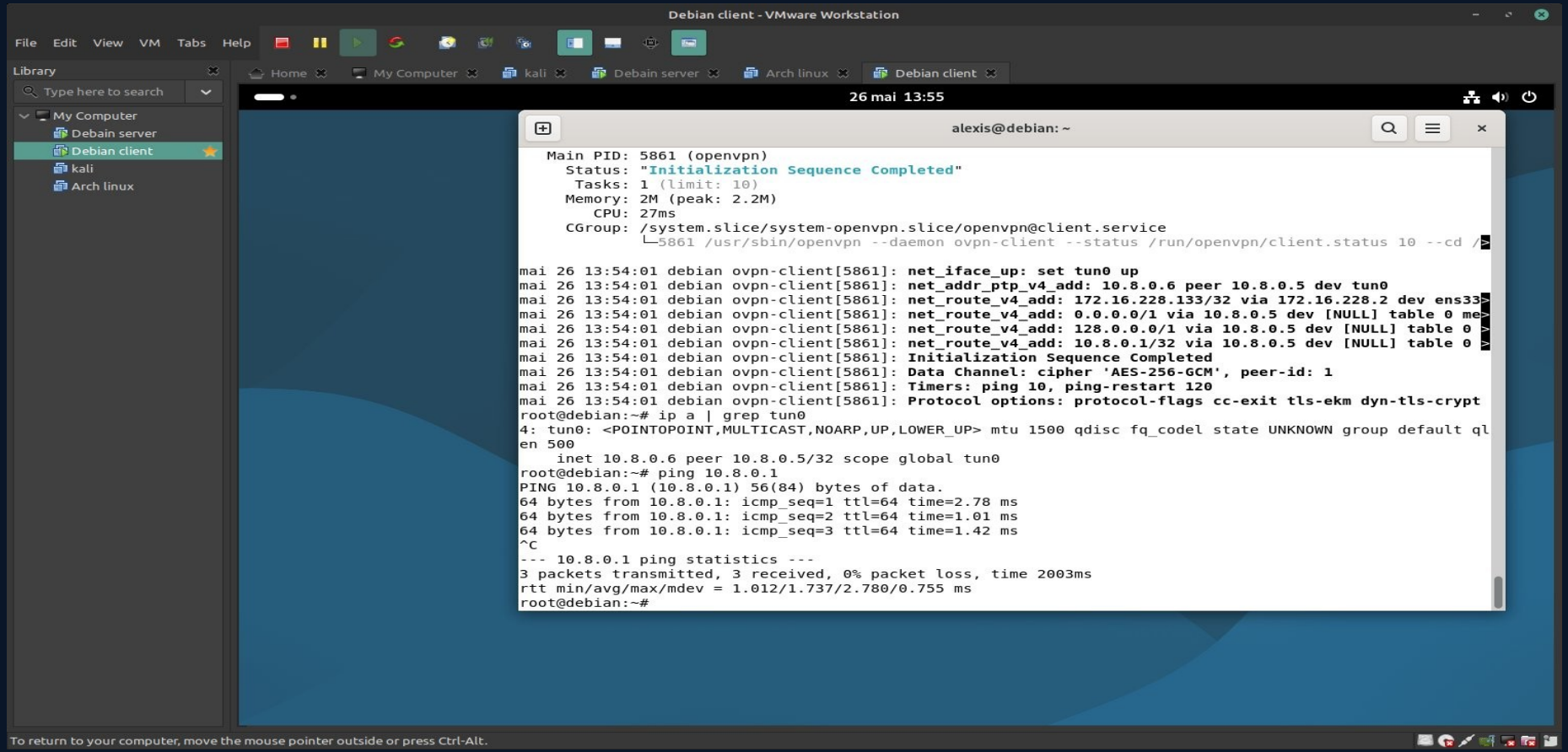
Initialization Sequence Completed — AES-256-GCM



Test — ✅

Ping 0% packet loss

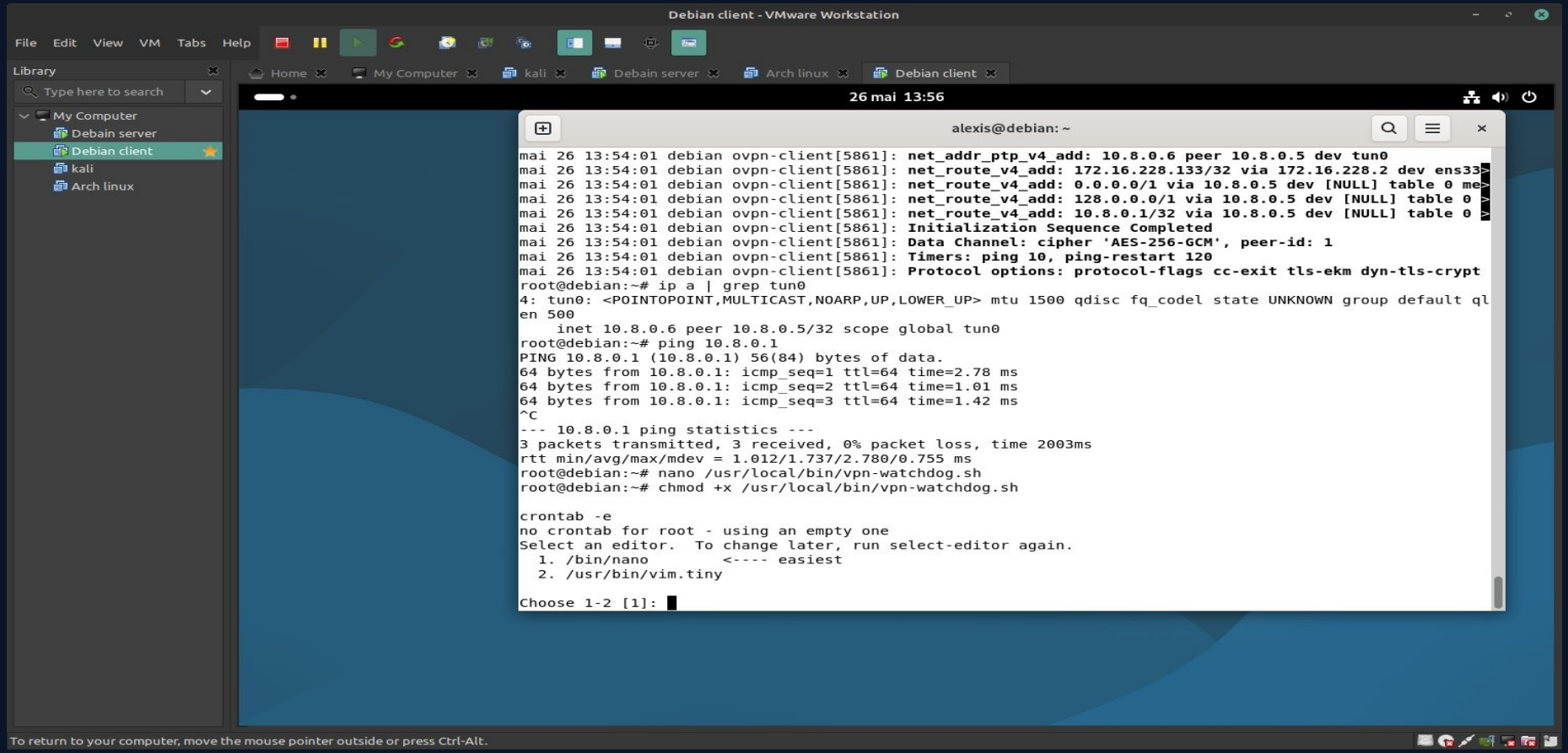
tun0: 10.8.0.6 — ping 10.8.0.1 → 3 paquets, 0% loss



Résilience — Étape 14

Watchdog — Choix éditeur cron

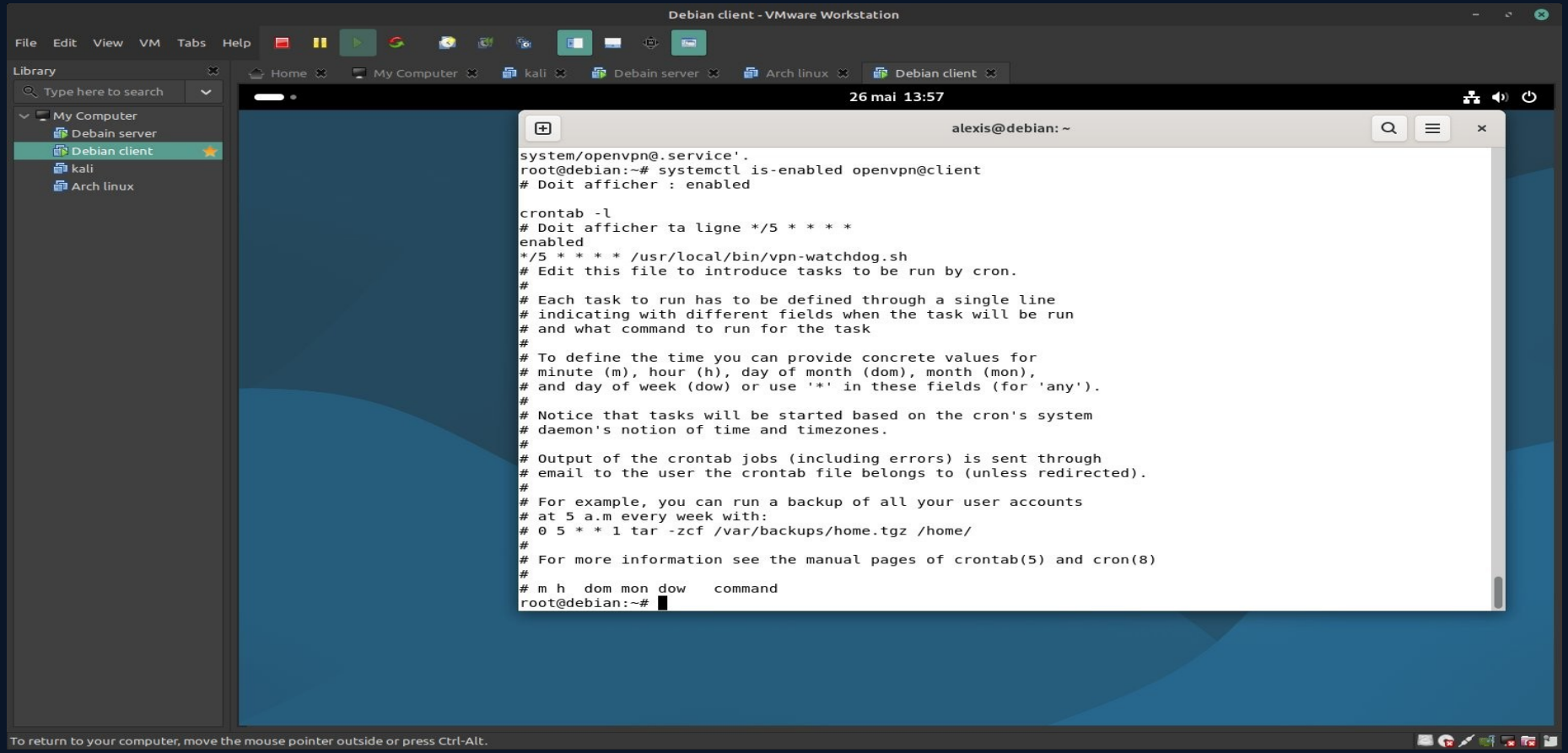
Script vpn-watchdog.sh créé et rendu exécutable



Résilience — ✅

Résilience configurée

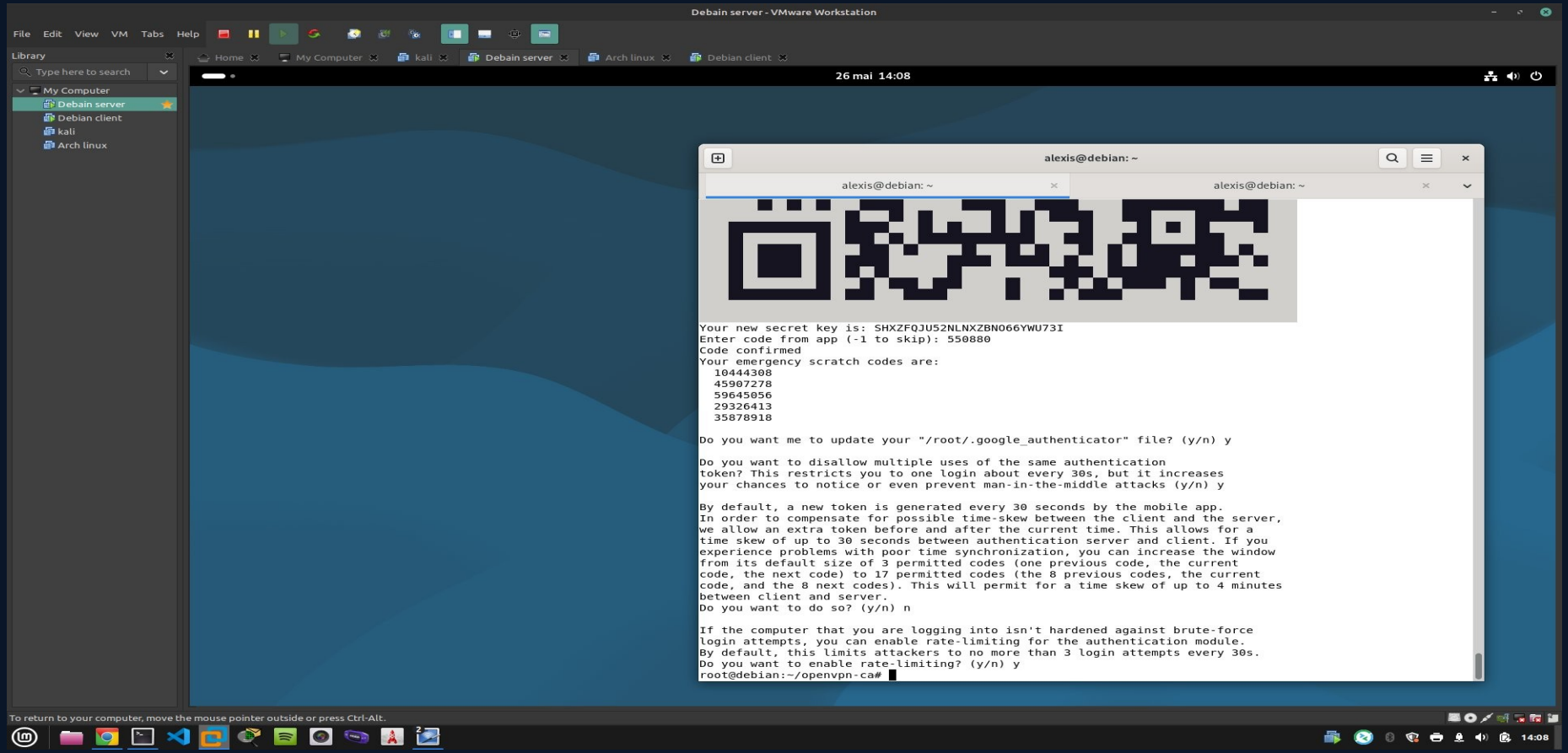
enabled + crontab */5 * * * * vpn-watchdog.sh



Bonus 2FA — Étape 1

Google Authenticator — QR Code

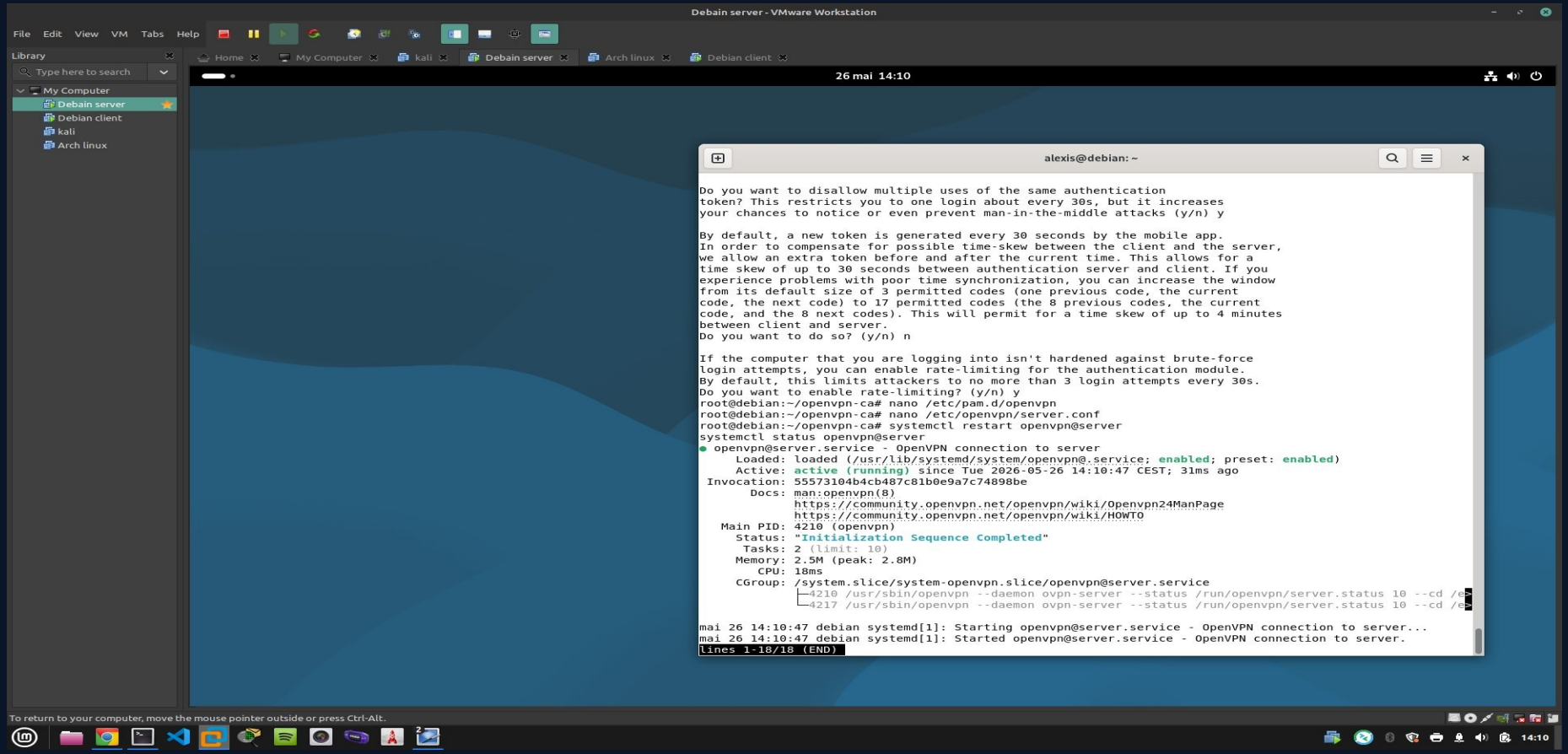
Scan du QR code — Code confirmed



Bonus 2FA — Étape 2

Serveur avec PAM activé

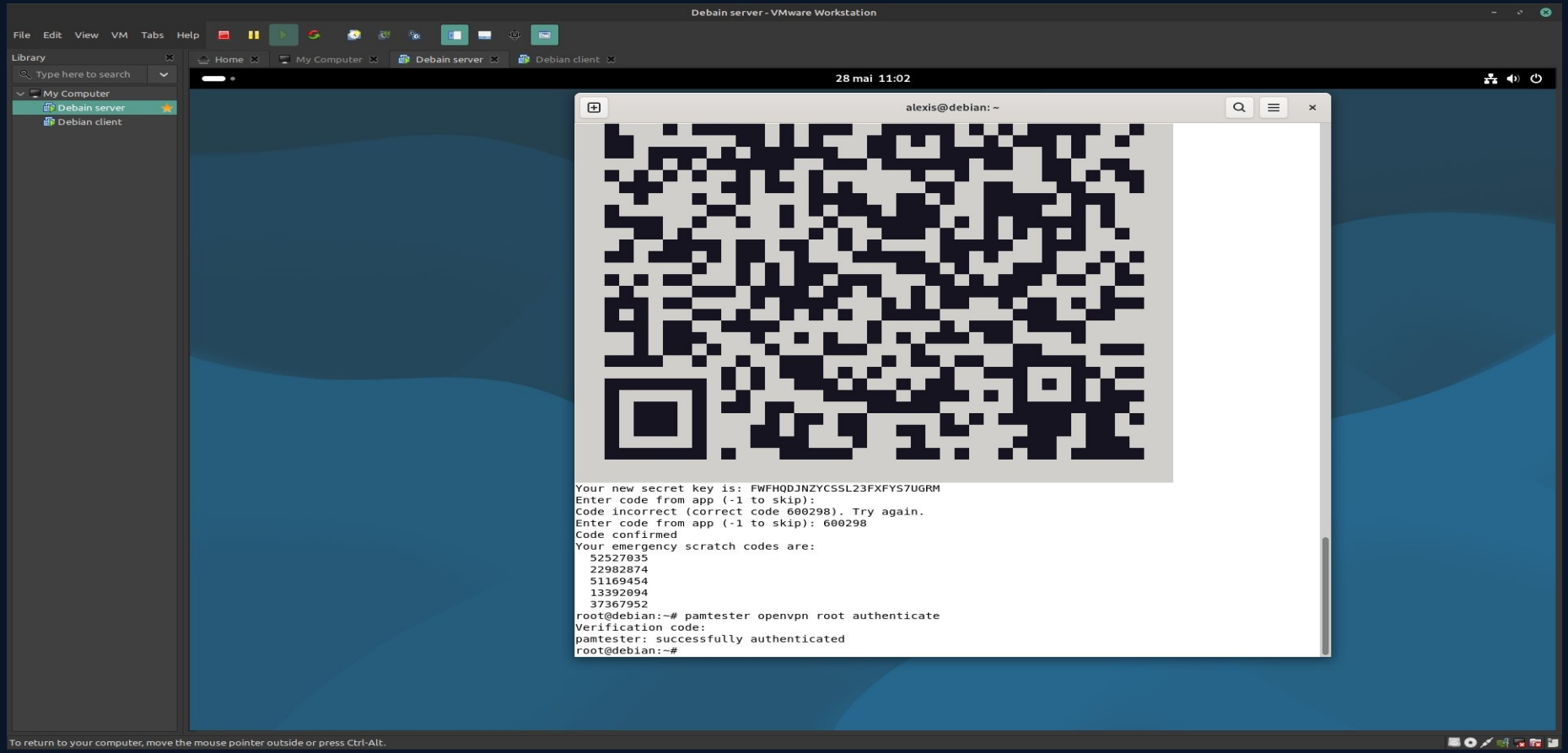
Plugin PAM chargé — Initialization Sequence Completed



Bonus 2FA — ✅ PAM OK

PAM — Authentification réussie

pamtester: successfully authenticated



Bilan & Compétences Acquises

- ✓ Infrastructure PKI complète (CA, certs serveur & client)
- ✓ Tunnel chiffré AES-256-GCM opérationnel
- ✓ Authentification mutuelle TLS (tls-auth)
- ✓ Forwarding IP + routage via le tunnel
- ✓ Démarrage automatique (systemd enable)
- ✓ Watchdog de résilience (cron + script)
- ✓ PAM Google Authenticator configuré (2FA partiel)

Compétences visées

→ Administrer les infras systèmes

→ Sécuriser les infras réseaux

→ PKI & gestion de certificats

→ Politique de sécurité VPN

→ Résilience & monitoring

→ OpenVPN, IPsec, AES, TLS